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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

3881145647

BIOLOGY **5090/23**

Paper 2 Theory **October/November 2010**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.
Section B
Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.
Write your answers in the spaces provided on the Question Paper.
Write an **E** (for Either) or an **O** (for Or) next to the number 8 in the Examiner's grid below to indicate which question you have answered.

You are advised to spend no longer than one hour on Section A and no longer than 45 minutes on Section B.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
Section B	
6	
7	
8	
Total	

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UNIVERSITY of CAMBRIDGE International Examinations

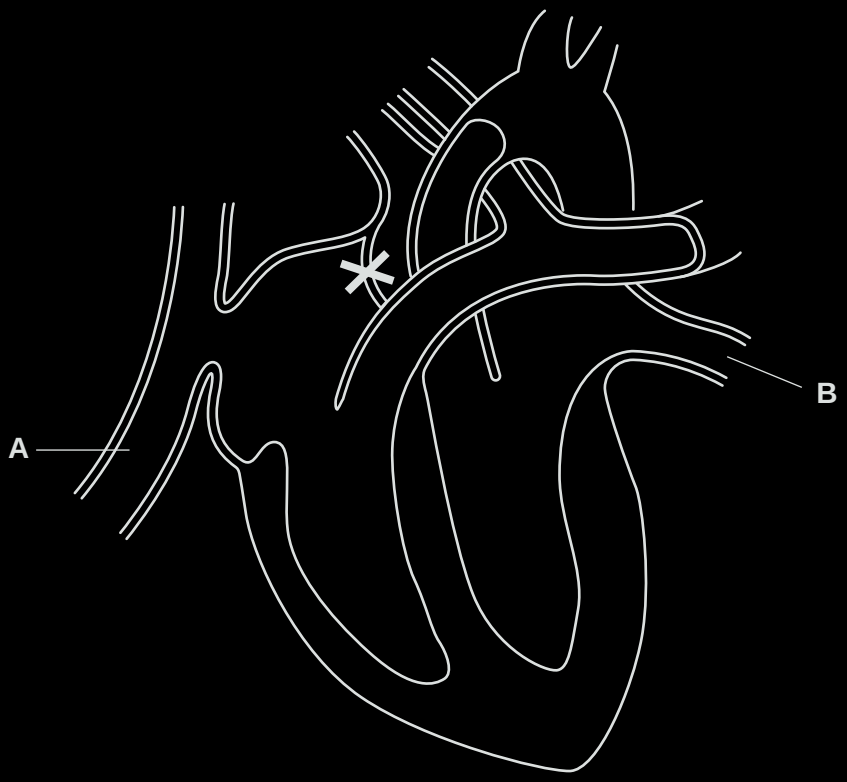


Fig. 1.1

(a) Identify structures **A** and **B** in Fig. 1.1.

A .....

B ..... [2]

(b) Complete Fig. 1.1 by drawing the valves to show how they control the direction of blood flow through the heart. [4]

(c) On Fig. 1.1, draw arrows to show the direction of blood flow into, through and out of the heart. [2]

Some people have a condition known as ‘a hole in the heart’. This allows a connection between the left and right atrium at point **X** in Fig. 1.1.

(d) Suggest two problems this might cause.

1. .....

2. .....[2]

[Total: 10]

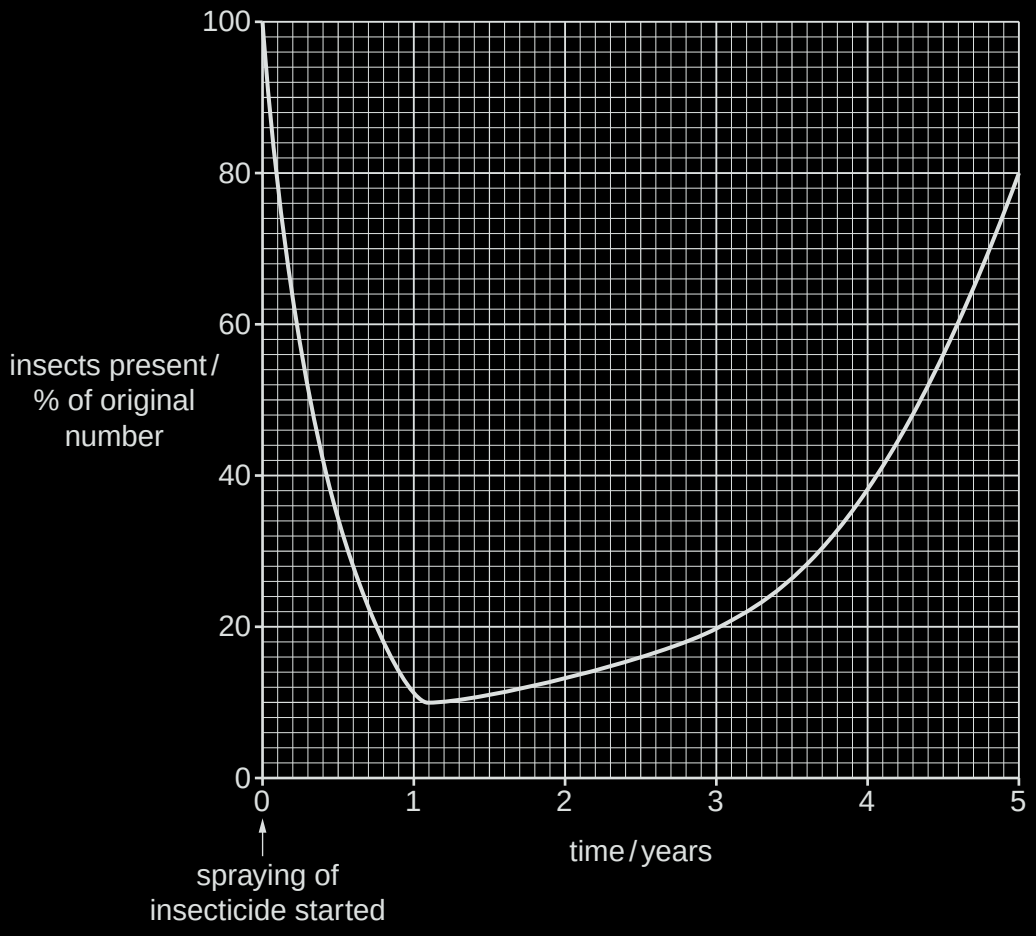


Fig. 2.1

(a) State **one** way an insect might be

(i) helpful to the crops .....

(ii) harmful to the crops. .....[2]

(b) Explain why the percentage of insects began to rise in the second year until almost reaching its original level.

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.....[5]

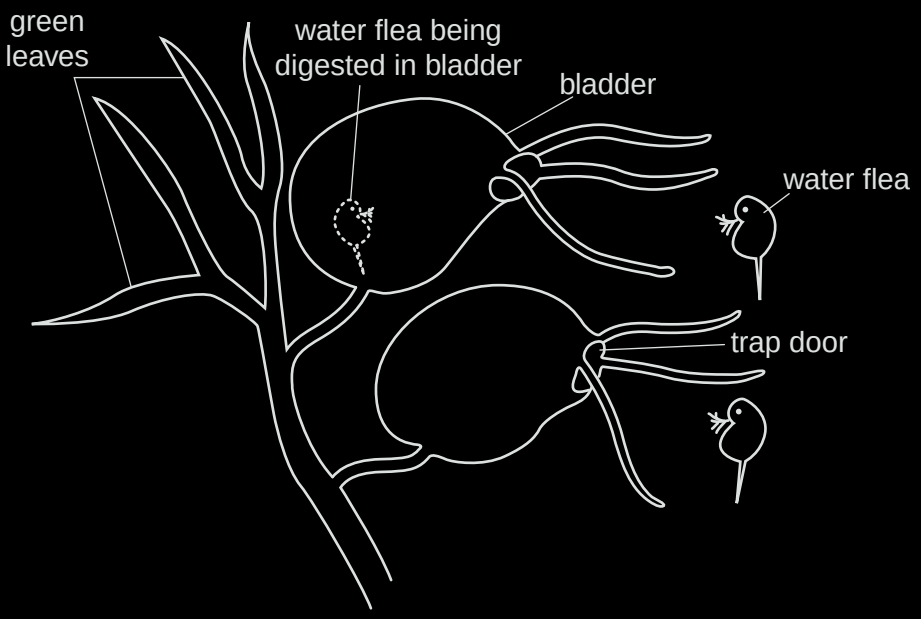


Fig. 3.1

(a) (i) State the process by which the plant is likely to obtain its carbohydrates.

.....[1]

(ii) Suggest how the raw materials for this process are made available to the plant.

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.....[4]

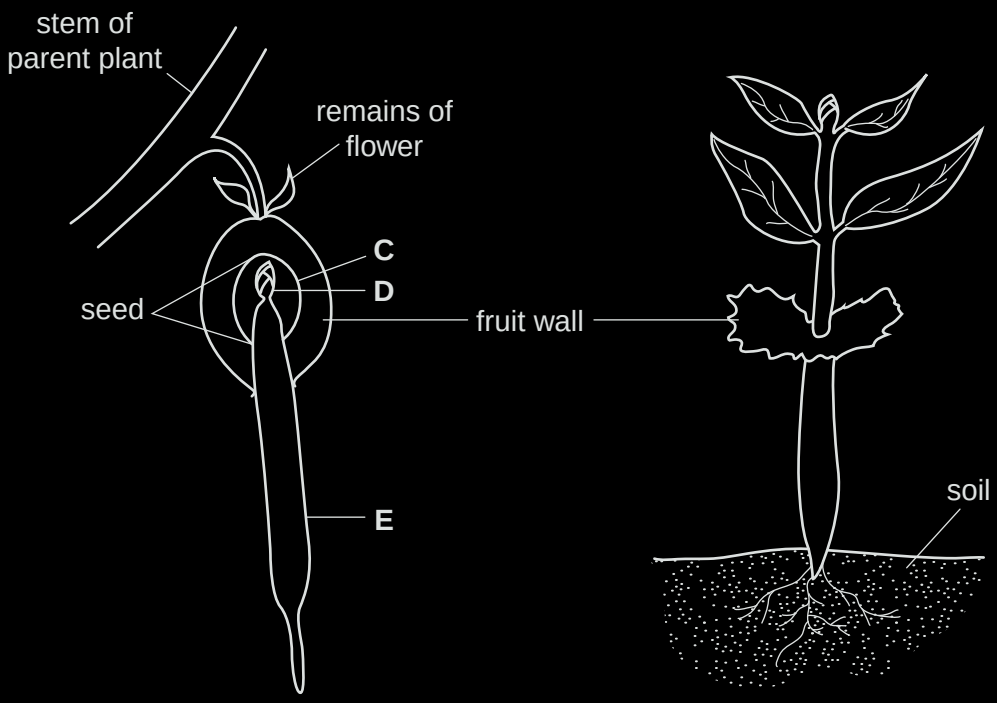


Fig. 4.1(a) **Fig. 4.1(b)**

(a) **(i)** Identify the structures **C** and **D** in Fig. 4.1(a).

C .....

D ..... [2]

(ii) State the part of the embryo from which structure **E** has developed.

.....[1]

(b) The air temperature is suitable for germination of the seed on the parent plant. Suggest how the embryo obtains the **other** requirements for germination.

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.....[4]

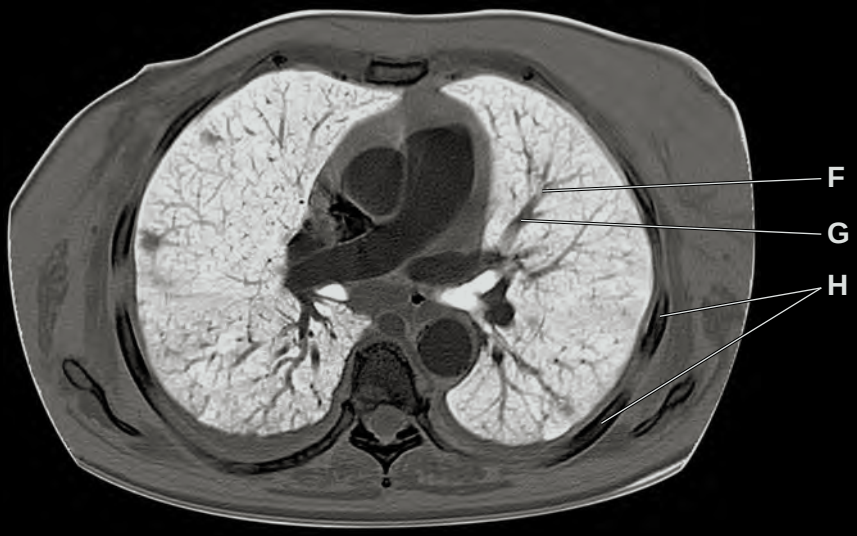


Fig. 5.1

(a) Draw a line on Fig. 5.1 to show the position of the chest wall if the person breathes in deeply. [1]

(b) Identify the structures **F**, **G**, **H** in Fig. 5.1.

F .....

G .....

H ..... [3]

(c) Explain the part played by structures **H** when the person breathes in.

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.....[5]

[Total: 9]